



Predicting SpaceX Falcon 9 First-Stage Landing Success

Data Science Capstone Project Report — IBM Applied Data Science Capstone

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Project GitHub repository: <https://github.com/keming-jiao/spacex-capstone>

● Outline

- 1 Executive Summary
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● Executive Summary

Methods

- Data collection: SpaceX REST API (v4) + web scraping of the Wikipedia Falcon 9 launch table
- Data wrangling: filtered to Falcon 9, imputed missing payload mass, built binary landing label Class
- EDA with Python visualisation (Matplotlib/Seaborn) and SQL (SQLite)
- Interactive visual analytics: Folium map and Plotly Dash dashboard
- Predictive analysis: Logistic Regression, SVM, Decision Tree and KNN tuned with 10-fold GridSearchCV

Key results

- 90 Falcon 9 launches analysed; overall landing success rate 66.7%
- Success rose from 0% (2010–2013) to 90% in 2019 — strong learning curve
- KSC LC-39A has the most successful launches (10 of 13, 76.9%)
- ES-L1, GEO, HEO and SSO orbits had 100% landing success; GTO was hardest (52%)
- Best model: SVM — 83.3% test accuracy, 84.8% CV accuracy
- Models never miss a real landing (0 false negatives); the remaining error is false positives

● Introduction

Project background and context

- SpaceX advertises Falcon 9 launches at about \$62 million, while other providers charge \$165 million or more
- Most of that saving comes from recovering and reusing the first stage
- If we can predict whether the first stage will land, we can estimate the cost of a launch
- A new competitor (Space Y) can use this to price its bids against SpaceX

Problems we want to answer

- Problem statement: predict whether the Falcon 9 first stage will land successfully (Class = 1) or not (Class = 0)
- Which factors — payload mass, orbit, launch site, flight number, booster version — drive landing success?
- How has the success rate changed over time?
- Which classification model predicts landing outcome best, and how reliable is it?

● Methodology

Data collection → wrangling → EDA → interactive analytics → predictive modelling

● Methodology — Overview

Step	What was done
Data collection	SpaceX REST API + web scraping of Wikipedia (requests, BeautifulSoup)
Data wrangling	Filter Falcon 9, handle missing values, derive Class label from landing Outcome
EDA	Visualisation (scatter, bar, line charts) and SQL queries on SQLite
Interactive visual analytics	Folium maps (markers, clusters, distance lines) and Plotly Dash dashboard
Predictive analysis	Standardise, train/test split, GridSearchCV over LR, SVM, Tree, KNN; evaluate on test set

GitHub URL: <https://github.com/keming-jiao/spacex-capstone>

● Data Collection

Two complementary sources were combined to describe every Falcon 9 launch

Source 1 — SpaceX REST API (api.spacexdata.com/v4/launches/past)

- GET request → JSON → pandas DataFrame with `json_normalize`
- Extra API calls by ID for rocket, launchpad, payload and core details

Source 2 — Web scraping (Wikipedia: List of Falcon 9 and Falcon Heavy launches)

- requests + BeautifulSoup parse the launch HTML tables
- Fields: flight no., date, booster, launch site, payload, mass, orbit, customer, launch & landing outcome

Output

- API → 90 Falcon 9 launches (`dataset_part_1.csv`); Wikipedia → 121 launch records (`spacex_web_scraped.csv`)

GitHub URL: <https://github.com/keming-jiao/spacex-capstone>

● Data Collection – SpaceX API

Flowchart of the API calls



Key code phrases

```
response = requests.get("https://api.spacexdata.com/v4/launches/past")
data = pd.json_normalize(response.json())
data = data[data['cores'].map(len)==1] # single-core launches
getBoosterVersion(data); getLaunchSite(data); getPayloadData(data); getCoreData(data)
data_falcon9 = launch_df[launch_df['BoosterVersion']!='Falcon 1']
data_falcon9['PayloadMass'].fillna(data_falcon9['PayloadMass'].mean(), inplace=True)
```

GitHub URL: https://github.com/keming-jiao/spacex-capstone/blob/main/01_data_collection_api.ipynb

● Data Collection – Web Scraping

Flowchart of the scraping process



Key code phrases

```
static_url = 'https://en.wikipedia.org/w/index.php?title=List_of_Falcon_9_and_Falcon_Heavy_launches&oldid=1027686922'
soup = BeautifulSoup(requests.get(static_url).text, 'html.parser')
html_tables = soup.find_all('table'); first_launch_table = html_tables[2]
column_names = [extract_column_from_header(th) for th in first_launch_table.find_all('th')]
for table in soup.find_all('table', 'wikitable plainrowheaders collapsible'): ...
df = pd.DataFrame({k: pd.Series(v) for k, v in launch_dict.items()})
```

GitHub URL: https://github.com/keming-jiao/spacex-capstone/blob/main/02_data_collection_web scraping.ipynb

● Data Wrangling

Cleaning, processing and creating the training label



- Missing values: PayloadMass imputed with mean during collection; LandingPad left missing for 26 launches where no pad was used
- Landing outcomes such as 'True ASDS', 'True RTLS', 'True Ocean' = success (Class = 1)
- bad_outcomes = {False ASDS, False Ocean, False RTLS, None ASDS, None None} → Class = 0
- Resulting label: 90 launches, 66.7% success (df['Class'].mean())
- One-hot encoding of Orbit, LaunchSite, LandingPad, Serial for modelling → dataset_part_3.csv (80 features)

GitHub URL: https://github.com/keming-jiao/spacex-capstone/blob/main/03_data_wrangling.ipynb

● EDA with Data Visualization

Charts plotted and why

Chart	Variables	Purpose
Scatter (catplot)	Flight Number vs. Launch Site	Does experience at a site improve landing?
Scatter	Payload Mass vs. Launch Site	Are heavy payloads launched from specific sites, and do they land?
Bar chart	Success rate by Orbit	Which orbit types are easiest / hardest to recover from?
Scatter	Flight Number vs. Orbit	Is orbit success driven by experience?
Scatter	Payload Mass vs. Orbit	How payload interacts with orbit type
Line chart	Yearly average success rate	Trend of landing success over time

GitHub URL: https://github.com/keming-jiao/spacex-capstone/blob/main/05_eda_dataviz.ipynb

● EDA with SQL

Queries performed on SPACEXTABLE (SQLite via %sql / sqlite3)

- Distinct launch site names (SELECT DISTINCT)
- 5 records where launch site begins with 'CCA' (LIKE 'CCA%' LIMIT 5)
- Total payload mass carried for NASA (CRS) (SUM)
- Average payload mass of booster F9 v1.1 (AVG)
- Date of the first successful ground-pad landing (MIN)
- Boosters with drone-ship success and $4000 < \text{payload} < 6000$ kg (WHERE ... AND ...)
- Total number of successful and failed mission outcomes (GROUP BY)
- Boosters that carried the maximum payload (subquery with MAX)
- 2015 drone-ship failures by month (substr(Date,6,2))
- Rank landing outcomes between 2010-06-04 and 2017-03-20 (COUNT ... ORDER BY DESC)

GitHub URL: https://github.com/keming-jiao/spacex-capstone/blob/main/04_eda_sql.ipynb

● Interactive Visual Analytics – Folium Map & Plotly Dash

Folium map

- folium.Circle + Marker for every launch site (lat/long)
- MarkerCluster with green (Class = 1) / red (Class = 0) markers per launch
- MousePosition to read coordinates; PolyLine + distance label to the nearest coastline, railway, highway and city
- Why: see whether location and surroundings relate to success

Plotly Dash dashboard

- Dropdown: all sites or one launch site
- Pie chart: successful launches by site (all) or success vs. failure (one site)
- RangeSlider: payload range 0–10,000 kg
- Scatter: payload vs. outcome, coloured by booster version category
- Why: interactively compare sites, payload ranges and boosters

GitHub URL: https://github.com/keming-jiao/spacex-capstone/blob/main/06_launch_site_folium.ipynb

GitHub URL: https://github.com/keming-jiao/spacex-capstone/blob/main/07_spacex_dash_app.py

● Predictive Analysis (Classification)

How the models were built, evaluated, improved and selected



- Hyper-parameters tuned: LR (C), SVM (kernel, C, gamma), Decision Tree (criterion, splitter, max_depth, min_samples), KNN (n_neighbors, algorithm, p)
- Model improvement: grid search picks parameters by 10-fold cross-validation accuracy on the training set
- Evaluation: test-set accuracy, confusion matrix, comparison with a majority-class baseline
- Selection rule: highest test accuracy; ties broken by cross-validation accuracy

GitHub URL: https://github.com/keming-jiao/spacex-capstone/blob/main/08_machine_learning_prediction.ipynb

● Results

EDA results

Success rate grows with flight number and year; 66.7% overall. Orbit type and payload matter.

Interactive analytics

All sites sit near the coast; KSC LC-39A has the most successes; heavier payloads tie to newer boosters.

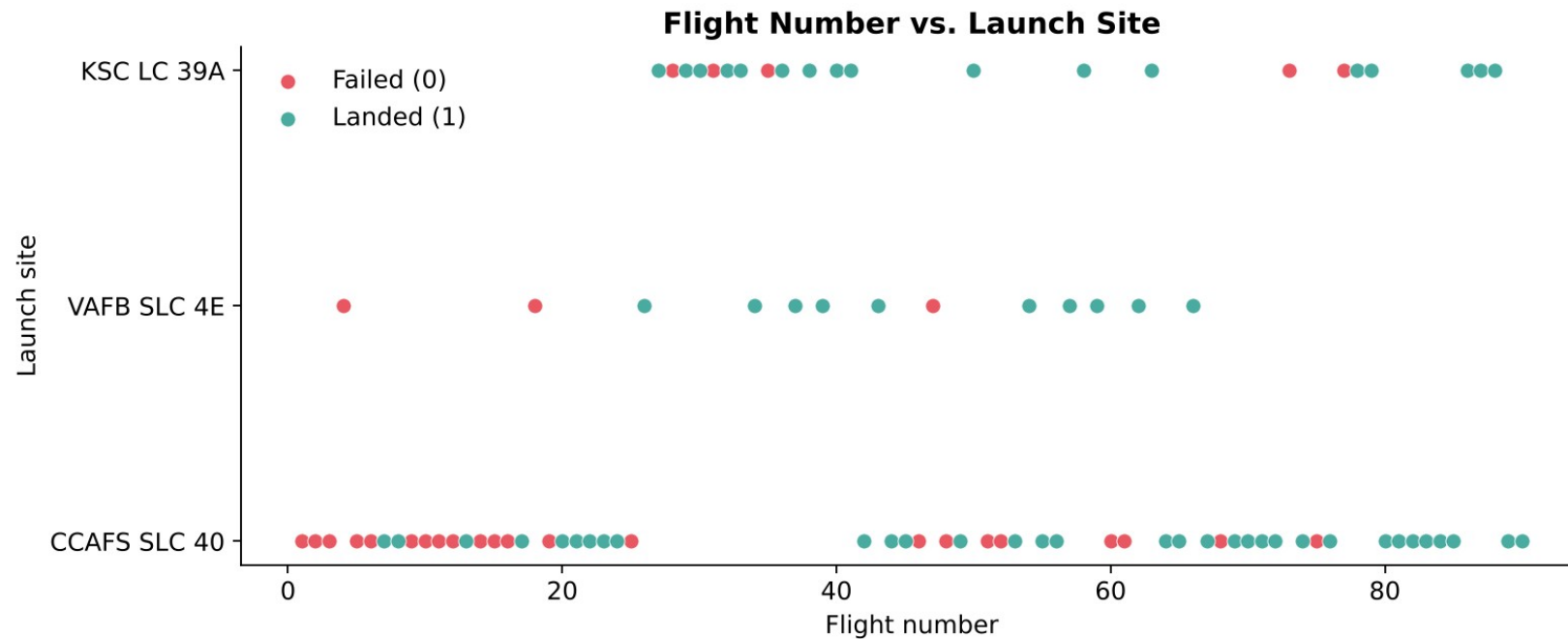
Predictive analysis

SVM best: 83.3% test accuracy; no false negatives.

● Insights drawn from EDA

Visualisation results

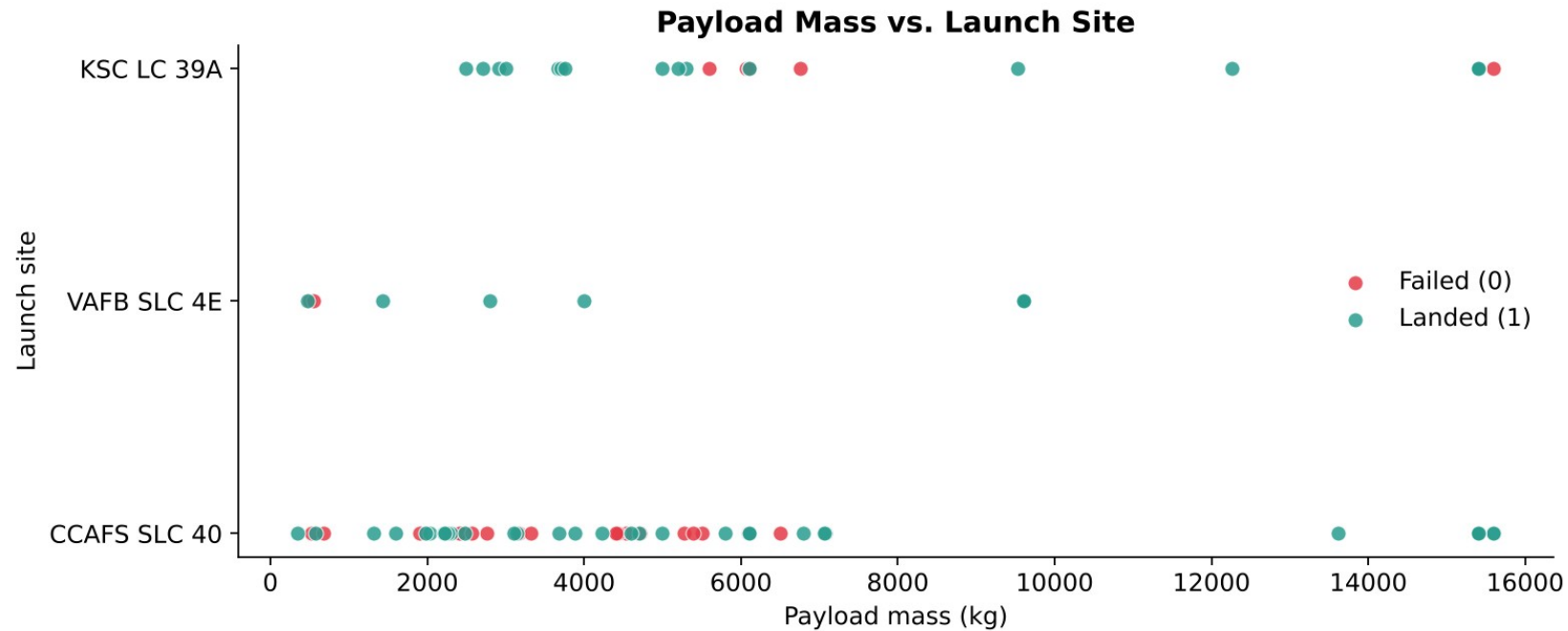
● Flight Number vs. Launch Site



Key takeaways

- Early flights (< 25) were mostly from CCAFS SLC 40 and mostly failed
- Later flights succeed far more often at every site — experience matters
- KSC LC 39A only starts after flight ~25, so it benefits from later, more mature boosters

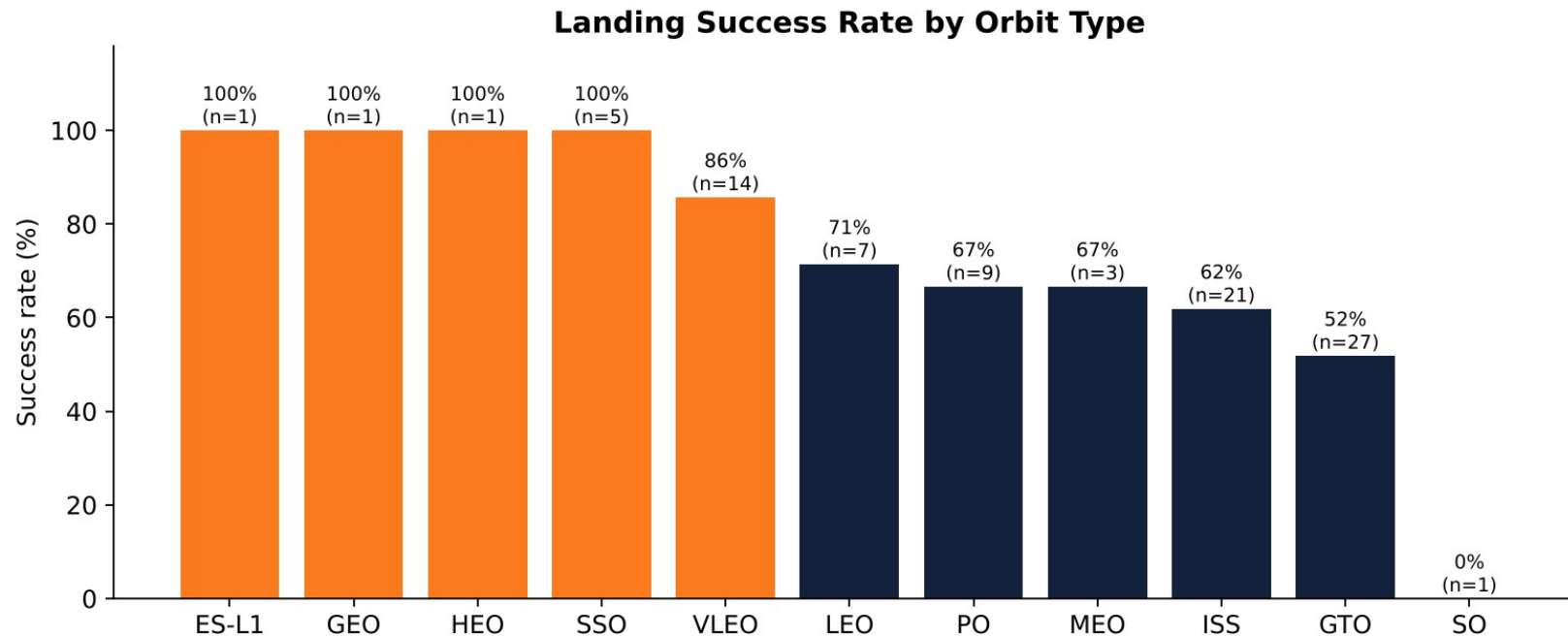
● Payload vs. Launch Site



Key takeaways

- VAFB SLC 4E launches no payload heavier than ~10,000 kg
- Heavy payloads (> 8,000 kg) mostly landed successfully
- Light payloads at CCAFS SLC 40 show many early failures

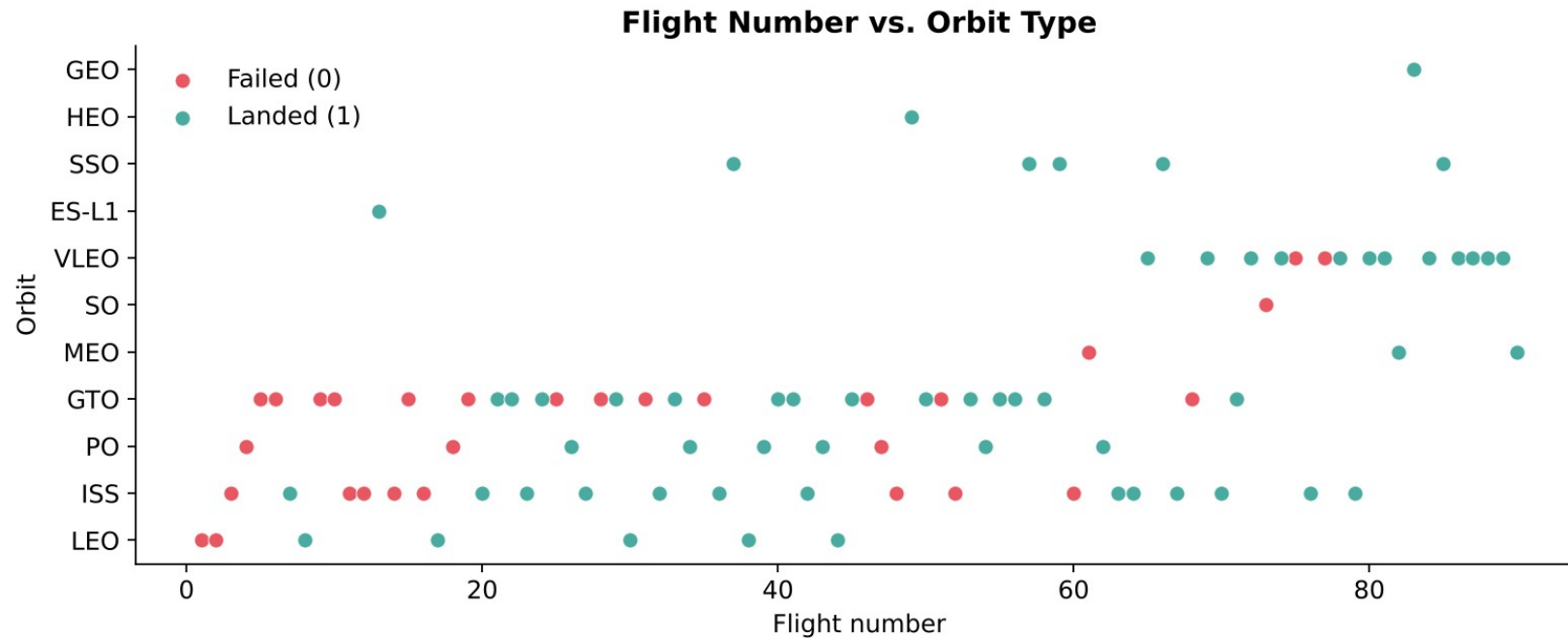
● Success Rate vs. Orbit Type



Key takeaways

- ES-L1, GEO, HEO and SSO: 100% success (but GEO/HEO/ES-L1 have only 1 launch each)
- VLEO 86% (n=14) is the most reliable orbit with a meaningful sample
- GTO is the hardest: 52% of 27 launches; SO 0% (n=1)

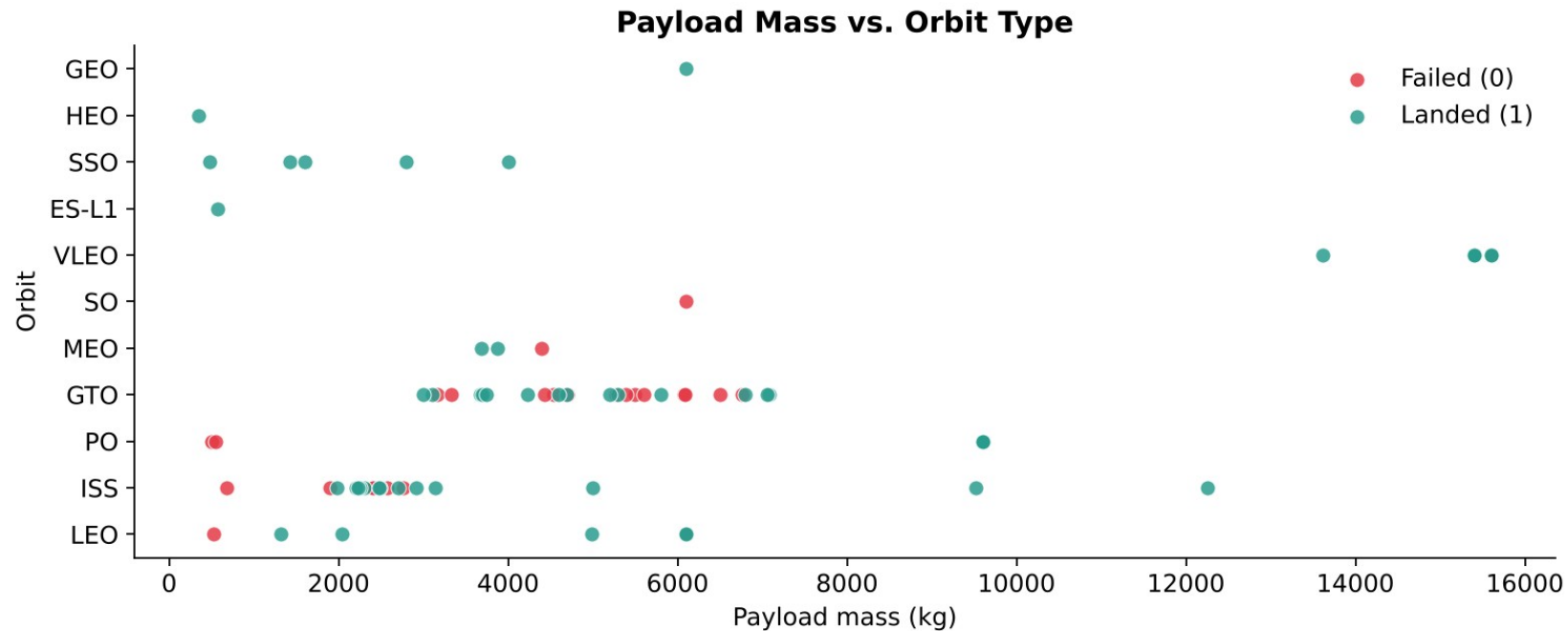
● Flight Number vs. Orbit Type



Key takeaways

- In LEO, success appears linked to the number of flights
- In GTO there is no clear relationship with flight number
- VLEO launches only began at high flight numbers and mostly succeed

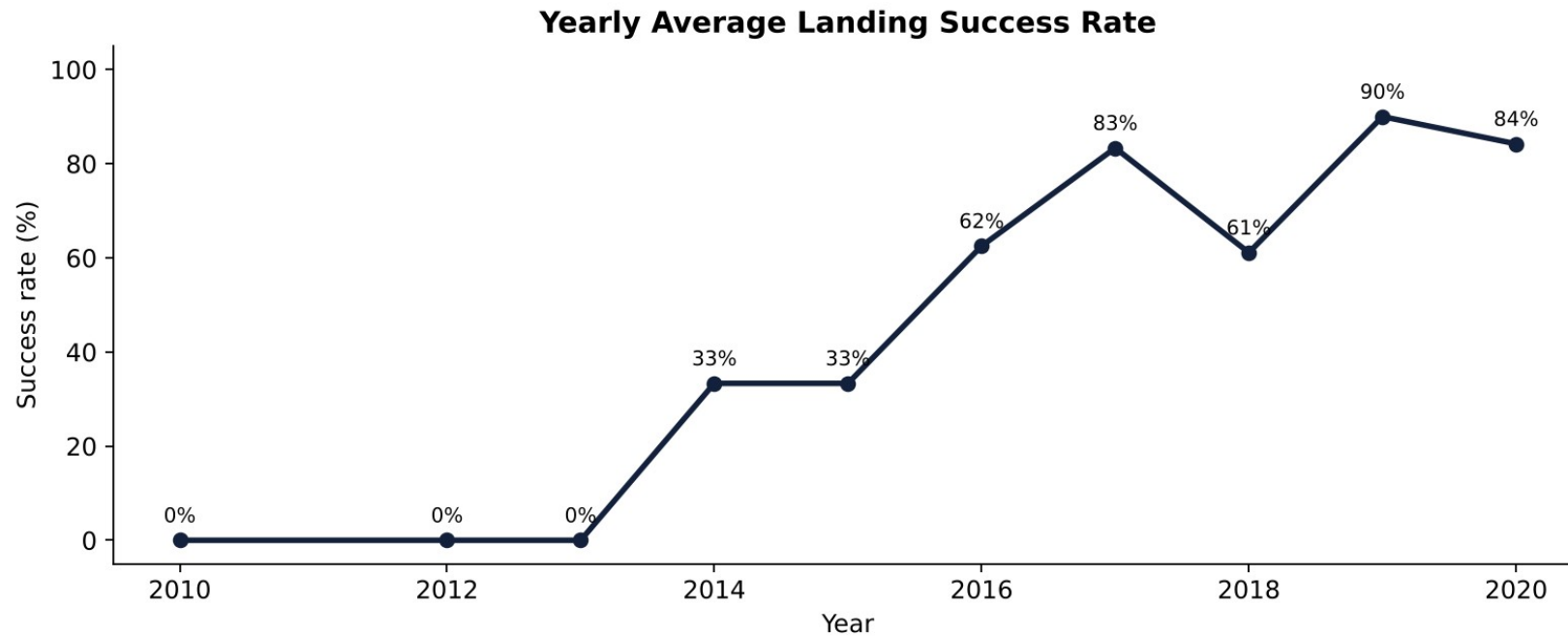
● Payload vs. Orbit Type



Key takeaways

- With heavy payloads, successful landings are more common for PO, LEO and ISS
- GTO mixes successes and failures across all payload masses
- Payload alone does not separate the classes — orbit must be considered too

● Launch Success Yearly Trend



Key takeaways

- 0% success from 2010 to 2013
- Success rate kept increasing from 2013 to 2020 (90% in 2019)
- Small dip in 2018 when new Block 4/5 hardware was introduced

● Insights drawn from EDA

SQL results

● All Launch Site Names

```
SELECT DISTINCT Launch_Site FROM SPACEXTABLE
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Result: SpaceX used 4 unique launch sites (CCAFS LC-40 and SLC-40 are the same pad renamed).

● Launch Site Names Beginning with 'CCA'

```
SELECT Date, Booster_Version, Launch_Site, Payload, Customer, Landing_Outcome FROM SPACEXTABLE WHERE Launch_Site LIKE 'CCA%' LIMIT 5
```

Date	Booster_Version	Launch_Site	Payload	Customer	Landing_Outcome
2010-06-04	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	SpaceX	Failure (parachute)
2010-12-08	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	NASA (COTS) NRO	Failure (parachute)
2012-05-22	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	NASA (COTS)	No attempt
2012-10-08	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	NASA (CRS)	No attempt
2013-03-01	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	NASA (CRS)	No attempt

Result: The first 5 launches were all from CCAFS LC-40; early landings failed or were not attempted.

● Total Payload Mass (NASA CRS)

```
SELECT SUM(PAYLOAD_MASS__KG_) AS total_payload FROM SPACEXTABLE WHERE Customer='NASA (CRS)'
```

total_payload
45596

Result: Boosters launched for NASA (CRS) carried 45,596 kg in total.

● Average Payload Mass by F9 v1.1

```
SELECT ROUND(AVG(PAYLOAD_MASS__KG_),2) AS avg_payload FROM SPACEXTABLE WHERE Booster_Version LIKE 'F9 v1.1%'
```

avg_payload
2534.67

Result: The F9 v1.1 booster family carried on average 2534.67 kg per launch.

● First Successful Ground Landing Date

```
SELECT MIN(Date) AS first_success FROM SPACEXTABLE WHERE Landing_Outcome='Success (ground pad)'
```

first_success
2015-12-22

Result: The first successful ground-pad landing happened on 2015-12-22.

● Successful Drone Ship Landing with Payload between 4000 and 6000

```
SELECT Booster_Version FROM SPACEXTABLE WHERE Landing_Outcome='Success (drone ship)' AND PAYLOAD_MASS__KG_>4000 AND PAYLOAD_MASS__KG_<6000
```

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Result: Four F9 FT boosters landed on a drone ship while carrying 4,000–6,000 kg.

● Total Number of Successful and Failure Mission Outcomes

```
SELECT Mission_Outcome, COUNT(*) AS n FROM SPACEXTABLE GROUP BY Mission_Outcome
```

Mission_Outcome	n
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Result: 100 of 101 missions succeeded (incl. 1 with unclear payload status); only 1 in-flight failure.

● Boosters Carried Maximum Payload

```
SELECT DISTINCT Booster_Version FROM SPACEXTABLE WHERE PAYLOAD_MASS__KG_=(SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTABLE)
```

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

Result: Only Block 5 (F9 B5) boosters carried the maximum payload (15,600 kg).

● 2015 Launch Records

```
SELECT substr(Date,6,2) AS month, Landing_Outcome, Booster_Version, Launch_Site FROM SPACEXTABLE WHERE  
substr(Date,1,4)='2015' AND Landing_Outcome='Failure (drone ship)'
```

month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Result: Two drone-ship landing failures in 2015 (January and April), both F9 v1.1 from CCAFS LC-40.

● Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

```
SELECT Landing_Outcome, COUNT(*) AS n FROM SPACEXTABLE WHERE Date BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY Landing_Outcome ORDER BY n DESC
```

Landing_Outcome	n
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

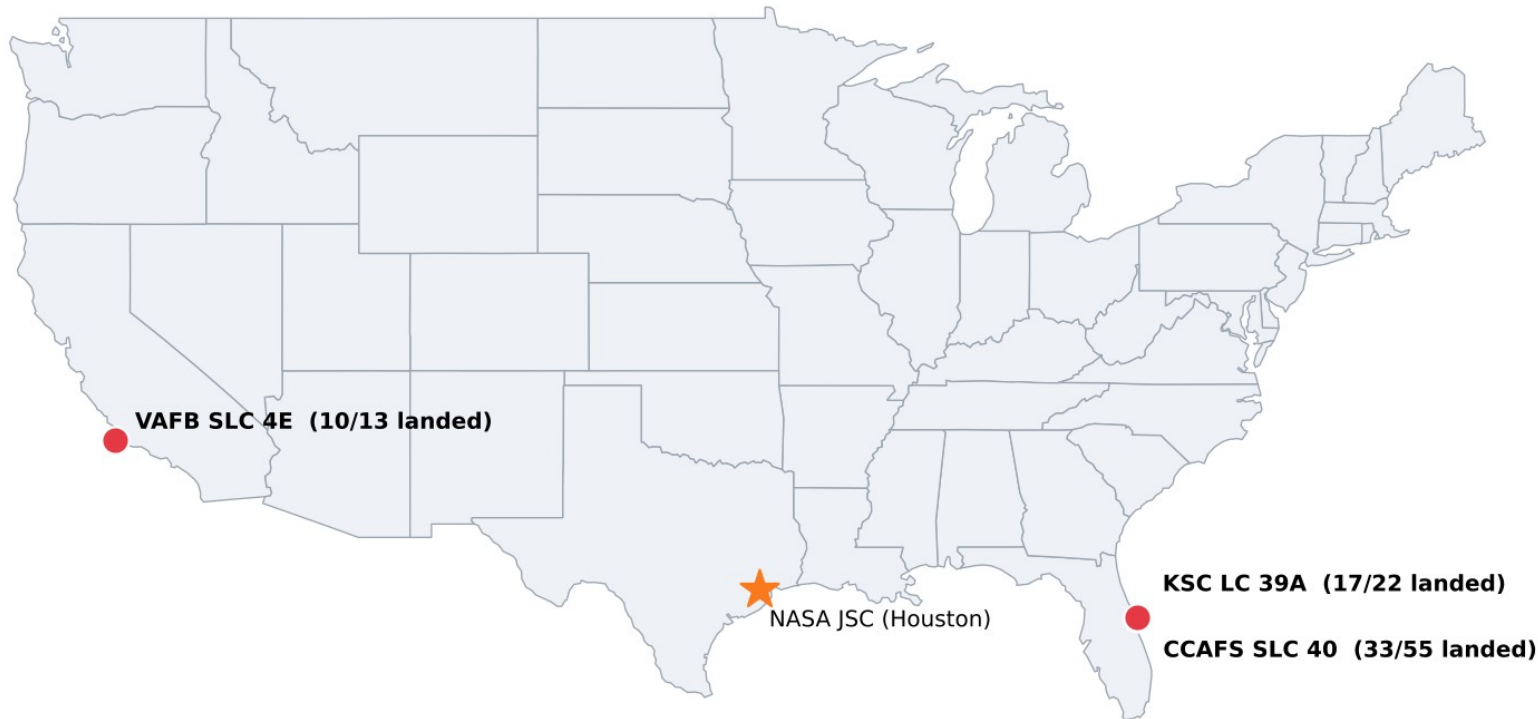
Result: 'No attempt' was most common; drone-ship successes and failures were tied at 5 each.

● Launch Sites Proximity Analysis

Folium map results

● All Launch Sites on a Global Map

All SpaceX Launch Sites (Folium Map)



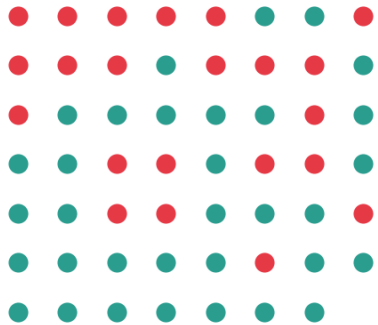
Key takeaways

- All launch sites are in the USA: Florida (east coast) and California (west coast)
- Every site is very close to the ocean — failed boosters fall into water, not on people
- Sites are as far south as possible to gain speed from Earth's rotation

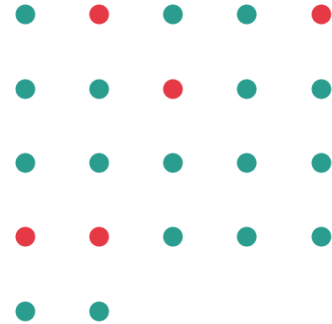
● Color-labeled Launch Outcomes per Site

Launch Outcomes per Site — green = success, red = failure (MarkerCluster)

CCAFS SLC 40
33/55 landed (60%)



KSC LC 39A
17/22 landed (77%)



VAFB SLC 4E
10/13 landed (77%)

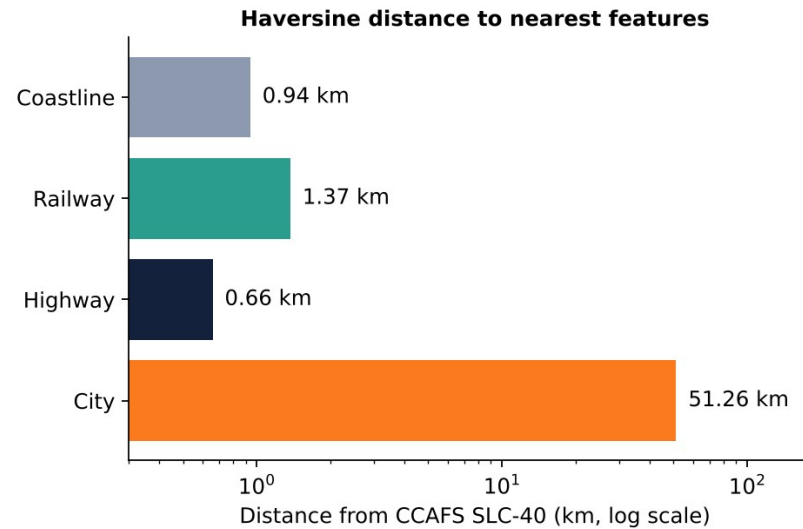
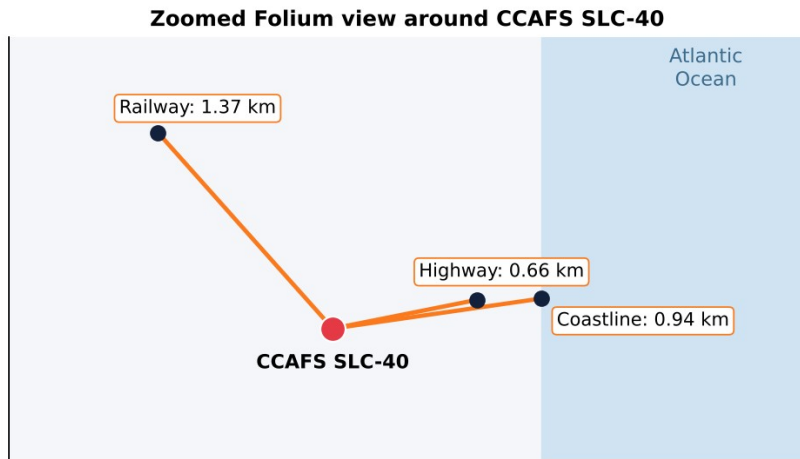


Key takeaways

- Green marker = successful landing, red = failure (MarkerCluster)
- KSC LC 39A and VAFB SLC 4E: about 77% success
- CCAFS SLC 40: most launches (55) but lowest rate (60%) because early test flights were there

● Proximity of a Launch Site to Its Surroundings

Proximity Analysis — MousePosition + folium.PolyLine distance lines



Key takeaways

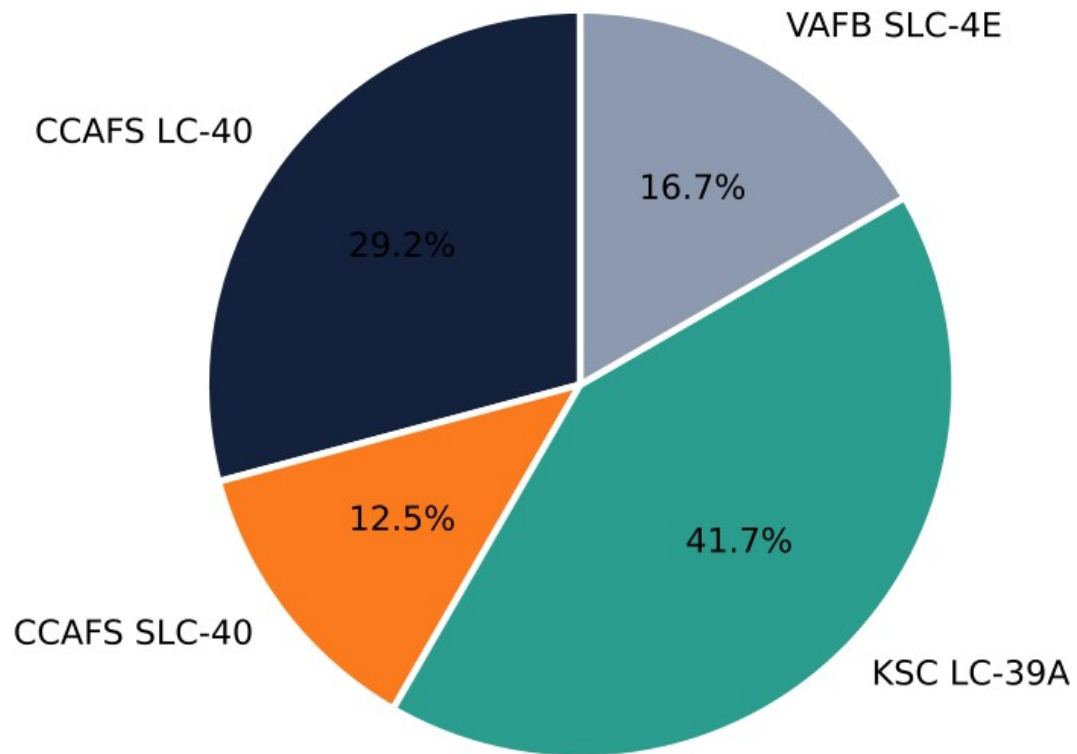
- Coastline 0.94 km, highway 0.66 km, railway 1.37 km — logistics are close
- Nearest city (Melbourne) is 51.26 km away — safety buffer from people
- Launch sites balance easy transport against public safety

● Build a Dashboard with Plotly Dash

Dashboard results

● Launch Success Count for All Sites

Total Successful Launches by Site (All Sites)

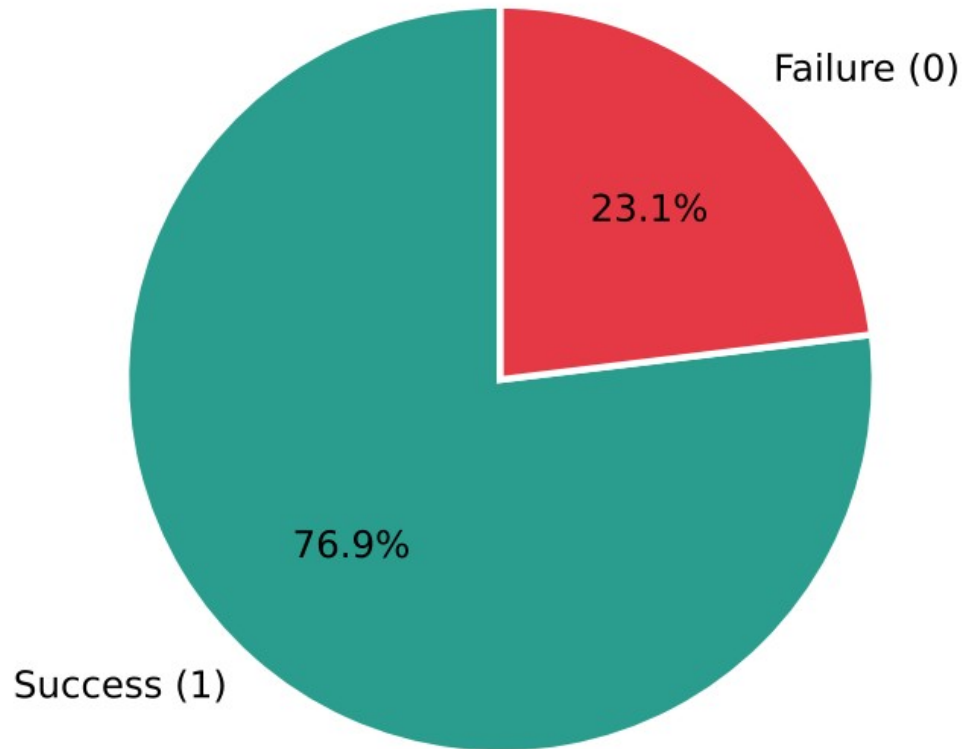


Key takeaways

- KSC LC-39A contributes the largest share of successful launches (10 of 24)
- CCAFS LC-40 is second; CCAFS SLC-40 and VAFB SLC-4E contribute least

● Launch Site with the Highest Success Ratio

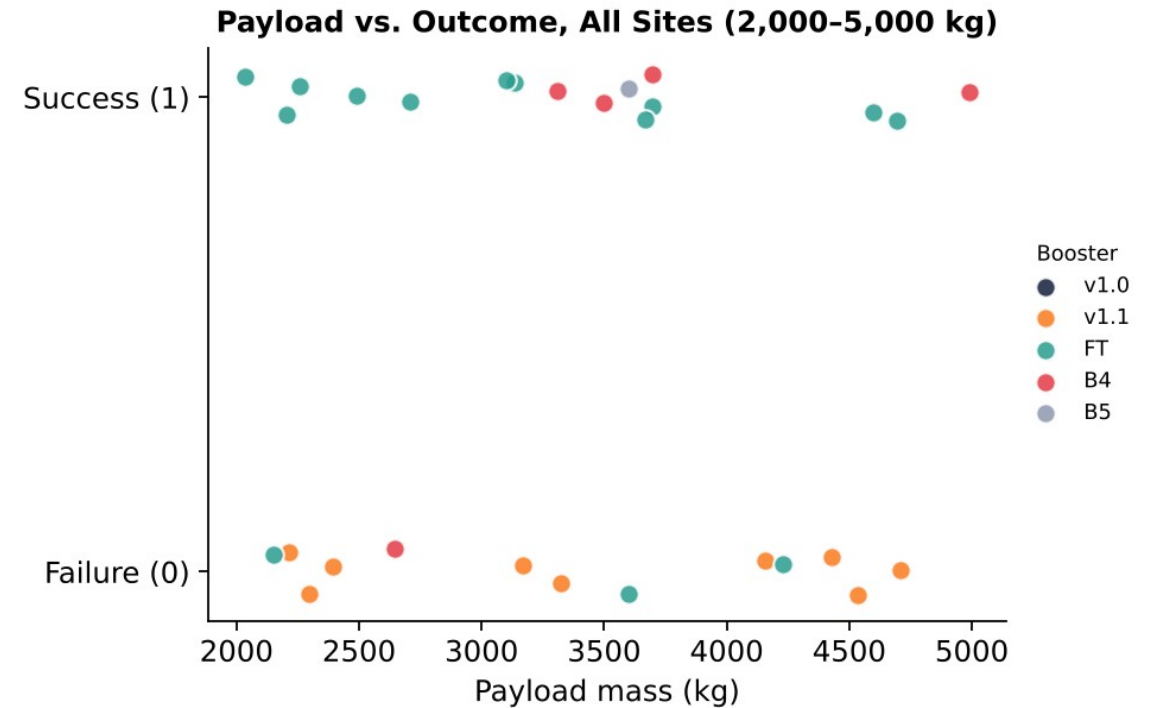
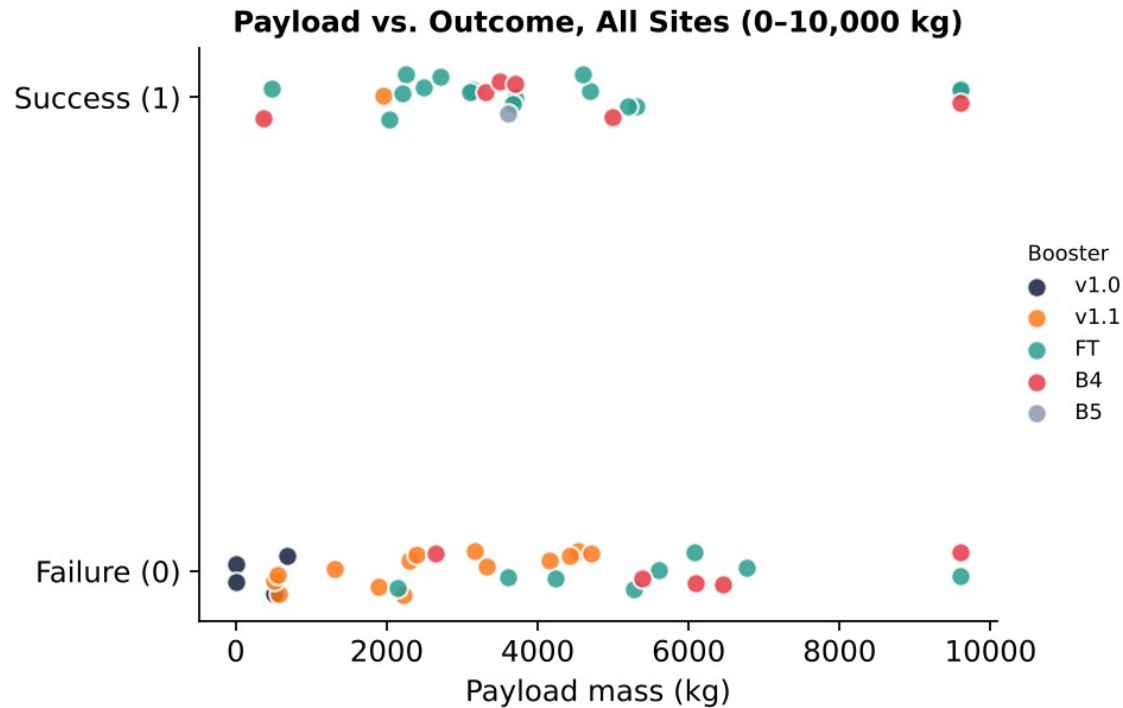
Success vs. Failure — KSC LC-39A



Key takeaways

- KSC LC-39A: 10 successes, 3 failures → 76.9% success
- Highest success ratio of all sites

● Payload vs. Launch Outcome for All Sites (Range Slider)

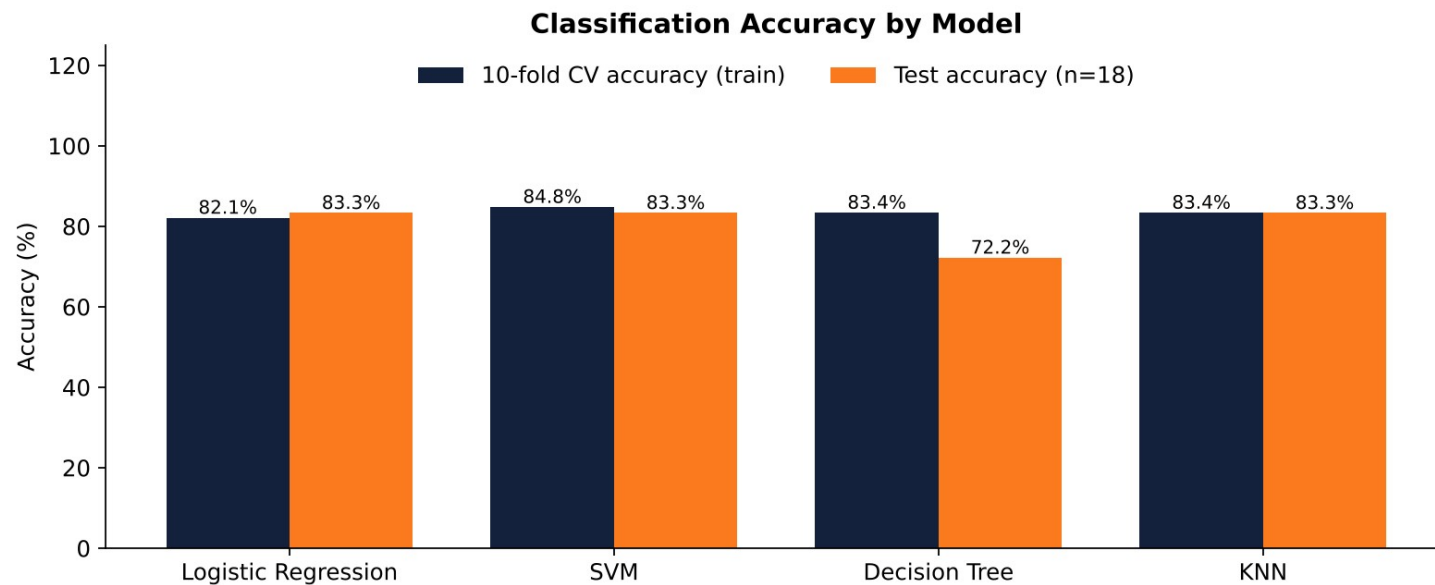


- Success rate for payloads $\leq 4,000$ kg is 47.1% vs. 36.4% above 4,000 kg
- FT boosters have the highest success rate among common versions (67%, n=24); v1.1 the lowest (7%, n=15)

● Predictive Analysis (Classification)

Model results

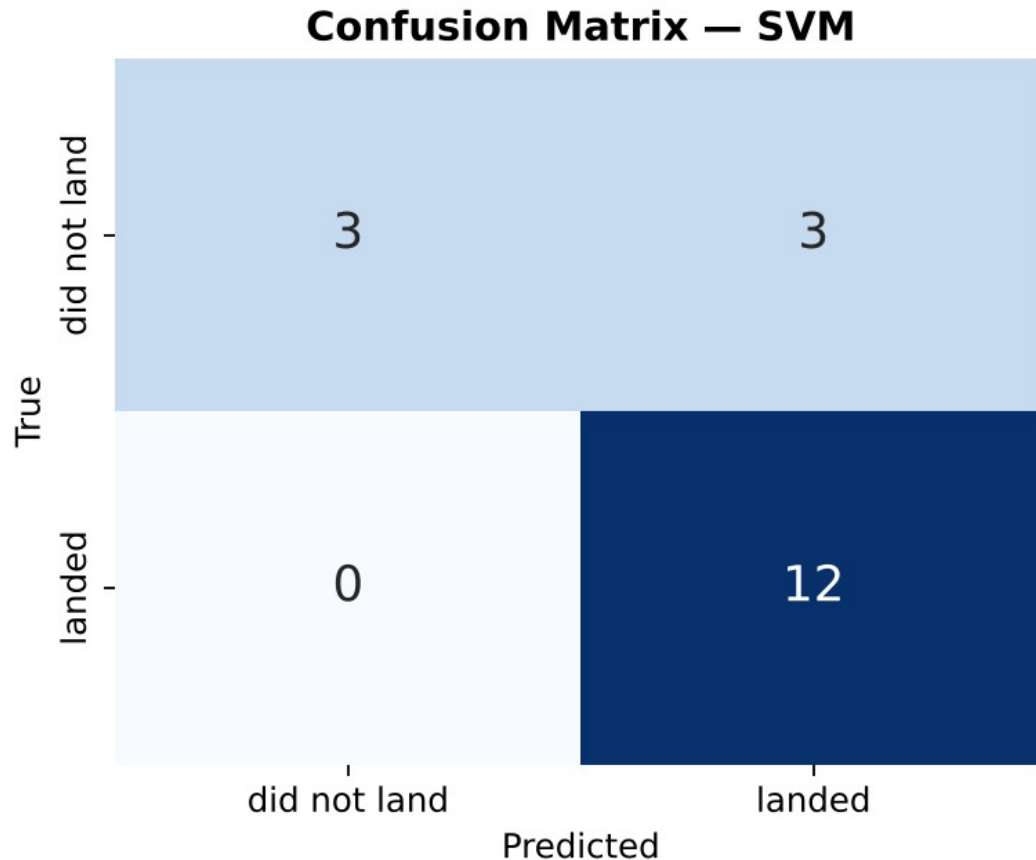
● Classification Accuracy



Model	CV acc.	Test acc.
Logistic Regression	82.1%	83.3%
SVM	84.8%	83.3%
Decision Tree	83.4%	72.2%
KNN	83.4%	83.3%

- LR, SVM and KNN tie on the test set (83.3%, 15/18 correct); Decision Tree drops to 72.2%
- SVM has the highest CV accuracy (84.8%) → best model
- Majority-class baseline is 66.7% — the models add ~17 points

● Confusion Matrix — Best Model (SVM)



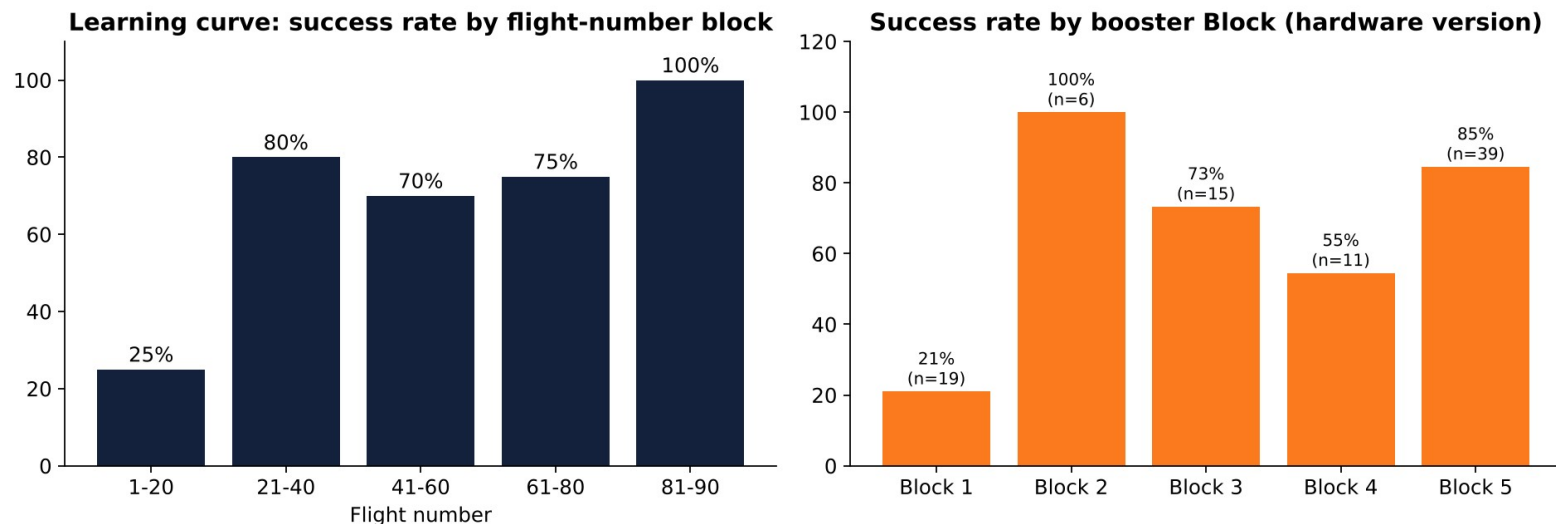
- True positives 12: every real landing in the test set was predicted correctly (recall = 100%)
- True negatives 3; false negatives 0
- False positives 3: the model sometimes predicts a landing that did not happen (precision = 80.0%)
- Business meaning: the model is optimistic — a bidder using it might under-estimate SpaceX's cost on risky missions
- Best parameters: kernel='sigmoid', C=1, gamma=0.0316

● Conclusions

- Landing success depends strongly on experience: from 0% before 2014 to 90% in 2019, and 100% for the last 10 flights
- Orbit matters: ES-L1, GEO, HEO, SSO and VLEO launches land most reliably; GTO is hardest
- KSC LC-39A has the most successful launches and the highest success ratio of any site
- Launch sites are close to coastline, highways and railways, but far from cities
- SVM is the best classifier (83.3% test, 84.8% CV accuracy); all models catch every real landing but over-predict success
- For Space Y: a launch to GTO with an older booster is the highest-risk scenario and should be priced with less reuse savings

● Innovative Insights — Beyond the Template

What actually drives a landing? Experience and hardware, not just payload



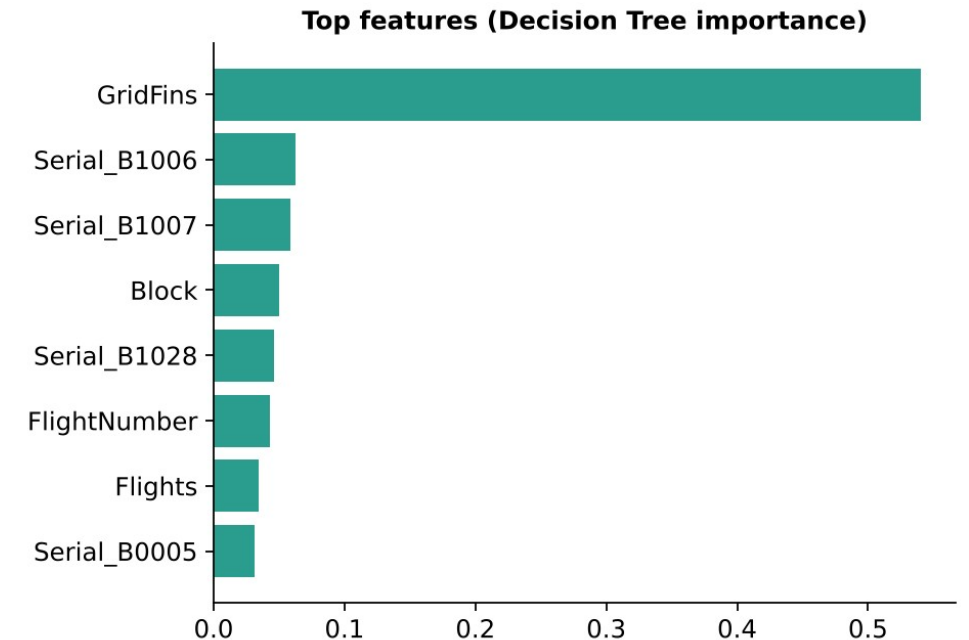
Idea for Space Y: price bids with a cost model = base cost – P(landing) × booster value, using P(landing) from the classifier.

- Learning curve: flights 1–20 landed only 25%; flights 81–90 landed 100%
- Block 1 boosters: 21%; Block 5: 85% (n=39)
- Reused boosters landed more often (78% vs. 58%) — reuse did not add risk
- GridFins is the single most important feature in the tree model (54% importance): it marks a real landing attempt

● Appendix

● Appendix — Hyper-parameters and Feature Importance

Model	Best parameters (GridSearchCV, cv=10)
Logistic Regression	C=1, penalty=l2, solver=lbfgs
SVM	C=1, gamma=0.0316, kernel=sigmoid
Decision Tree	criterion=gini, max_depth=10, max_features=sqrt, min_samples_leaf=1, min_samples_split=2, splitter=random
KNN	algorithm=auto, n_neighbors=6, p=1



Data files: dataset_part_1.csv (API), spacex_web_scraped.csv (scraping), dataset_part_2.csv (labelled), dataset_part_3.csv (features), SpaceX.csv (SQL), spacex_launch_dash.csv (dashboard)

GitHub URL: <https://github.com/keming-jiao/spacex-capstone>

Thank you!

<https://github.com/keming-jiao/spacex-capstone>